11 COMPONENTS OF AI PROJECT FRAMEWORK

Learning Outcomes

The students will be able to:

- know about the AI Project cycle and appreciate its importance.
- understand the nature of problem scoping.
- set goals for an AI project.
- explore the process of data acquisition.
- comprehend data exploration using data visualisation techniques.
- gain awareness about modelling and its approach.
- get familiar with evaluation and deployment.

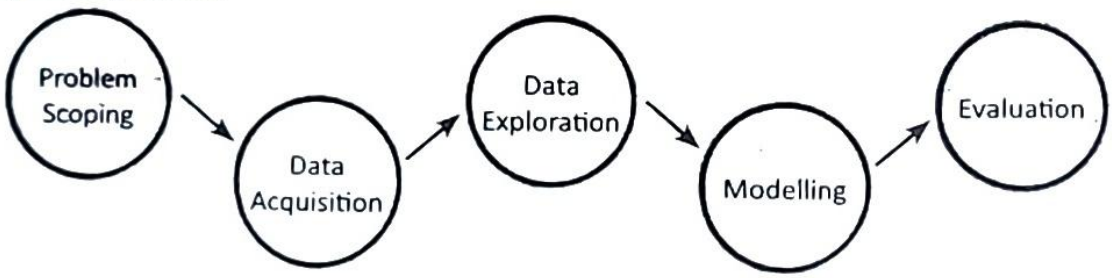
You must have studied a variety of cycles in your earlier grades that show the progression of numerous processes. For example, the water cycle shows how water continually travels both inside the Earth and through the atmosphere.

In the same way, the AI project cycle outlines the steps necessary for creating an AI project. It helps to understand the problems and come up with solutions for them.

While building the AI model, with the help of the AI project cycle, we define the problem, set goals, collect data, find patterns in the data, and test the performance of the AI model. The AI project cycle guides you through the process of developing AI systems. Let us learn about it in detail.

AI Project Cycle

The AI project cycle is a structured framework that outlines the different stages for creating an AI system. It is a systematic and step-by-step process of developing and implementing artificial intelligence projects. Each stage involves careful planning and execution to design and deploy AI solutions.



Components of AI Project Framework

Problem scoping : The first step is to understand and define the problem that we want AI to solve. Problem scoping is the stage where we set clear goals and outline the objectives of the AI project. It includes precisely outlining the issues, defining them explicitly, identifying their causes, and developing a plan to fix them.

Data acquisition : This stage focuses on collecting the relevant data required for the AI system.

Since the data forms the base of your project, care must be taken to ensure that the data is collected from reliable and authentic sources. Also, data should be in sufficient quantity to provide proper analysis.

Data exploration: This stage involves the exploration and analysis of the collected data to interpret patterns, trends, and relationships. The data is in large quantities. In order to easily understand the patterns, you can use different visual representations, such as graphs, databases, flowcharts, and maps.

Modelling: After exploring the patterns, you need to select the appropriate AI model to achieve the goal. This model should be able to learn from the data and make predictions.

Evaluation: The selected AI model now needs to be tested, and the results need to be compared with the expected outcome. This helps in evaluating the accuracy and reliability of the model and improving it.

Once the evaluation is complete and generates accurate results, the AI model can be deployed in the real world.

Activity 11.1

A. Consider the following real-life scenario to understand a practical approach for following the AI project cycle.

Suppose you have been appointed as the Chief Security Officer for the protection of high-value ancient artefacts that have been brought into a museum to be exhibited. These artefacts are of great historical significance, and collectors around the world want to include them in their collections.

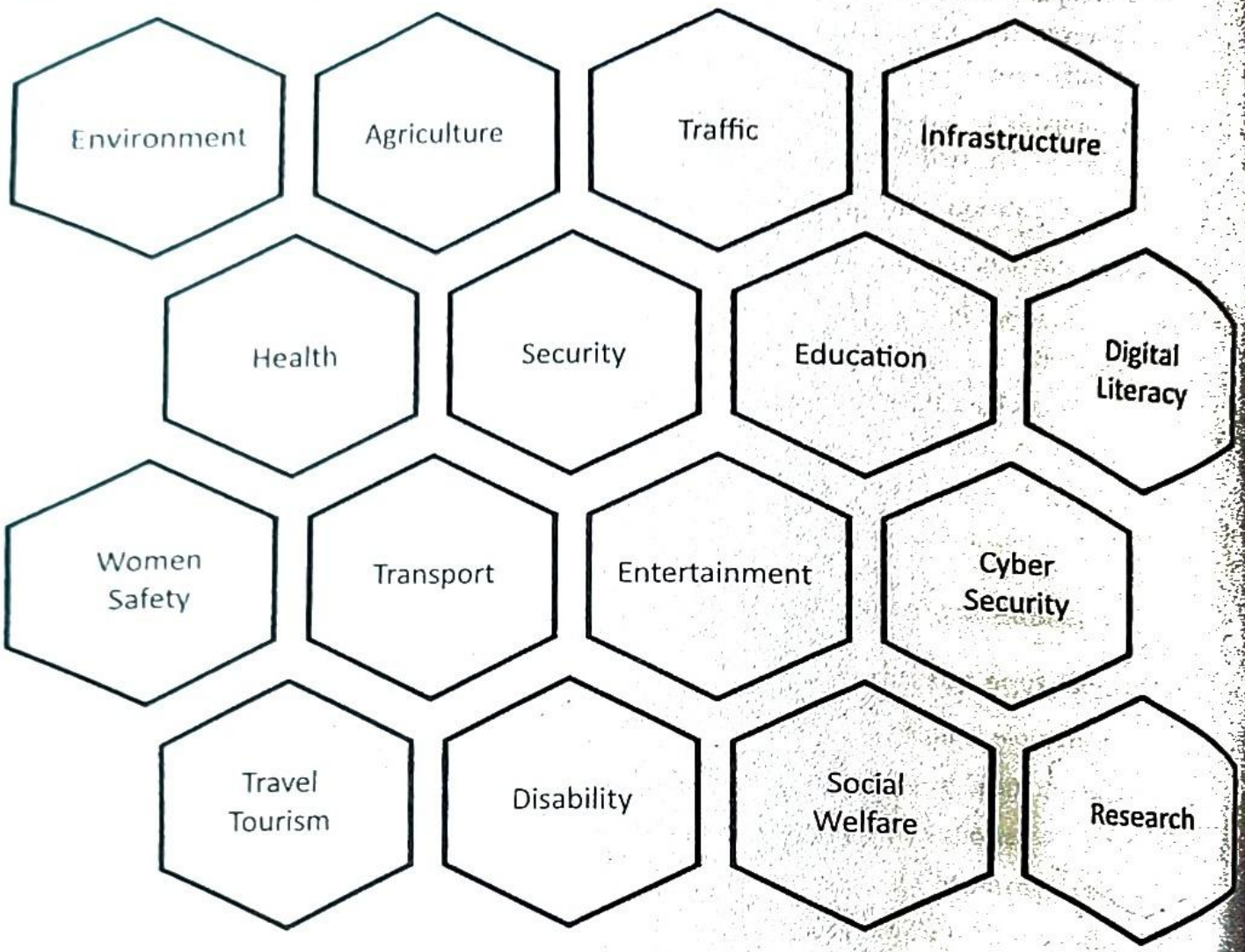


Some information is circulated that a notorious art collector has recruited a master criminal to steal the artefacts. Your task is to enhance the security of the museum and prevent anyone from stealing the precious artefacts.

B. Explain the concept of the AI project cycle with the help of a suitable example.

Problem Scoping and Setting Goals

As you have learnt, problem scoping refers to the process of choosing a problem that we w address using AI. It involves identifying the problem's scope, objectives, and constraints in orde establish a clear understanding of what needs to be addressed. In order to understand the pro scoping, let us select a problem from one of the themes given below:



For more options, you can either select an SDG-related issue around you that needs to be solv using AI or you can think of one on your own.

The following are some suggested themes:

Theme	Topics or Challenges Related to the Theme
Smart Cities	Energy efficiency, traffic management, waste management, etc.
Social Media	Sentiment analysis, fake news detection, content management, etc.
Environmental Sustainability	Biodiversity conservation, climate action, pollution control, etc.
Digital Literacy	Online learning platforms, digital awareness, e-books, etc.
Health	Medicinal aid, mobile medications, spreading of diseases, etc.
Entertainment	Media, virtual gaming, interactive advertisements, promotions, etc.
Agriculture	Pest issues, yield rates, sowing and harvesting patterns, etc.

Highway	Land acquisition, land quality, deforestation, road construction, etc.
Vaccination Drive	Vaccine awareness, limited vaccine supply, setting up vaccination centres, training healthcare workers, etc.
Hostel	Limited space and facilities, surveillance systems, maintenance and cleanliness, rent collection, responding to complaints, coordinating maintenance activities, etc.
Smart School Attendance System	Connectivity problems, hardware malfunctions, software bugs, user adoption and resistance, cost and maintenance, inaccurate attendance records, etc.

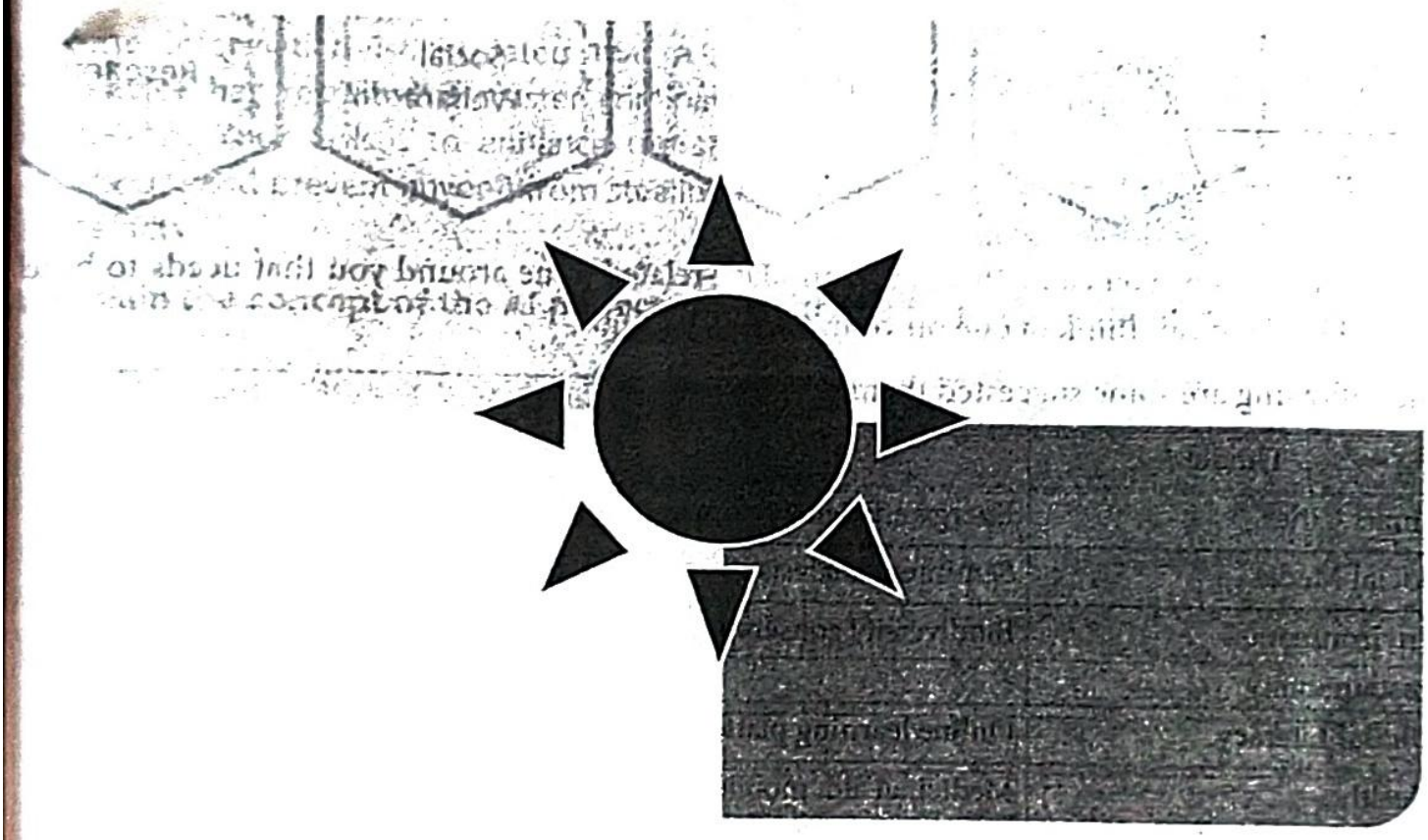
Activity 11.2

Answer the following questions.

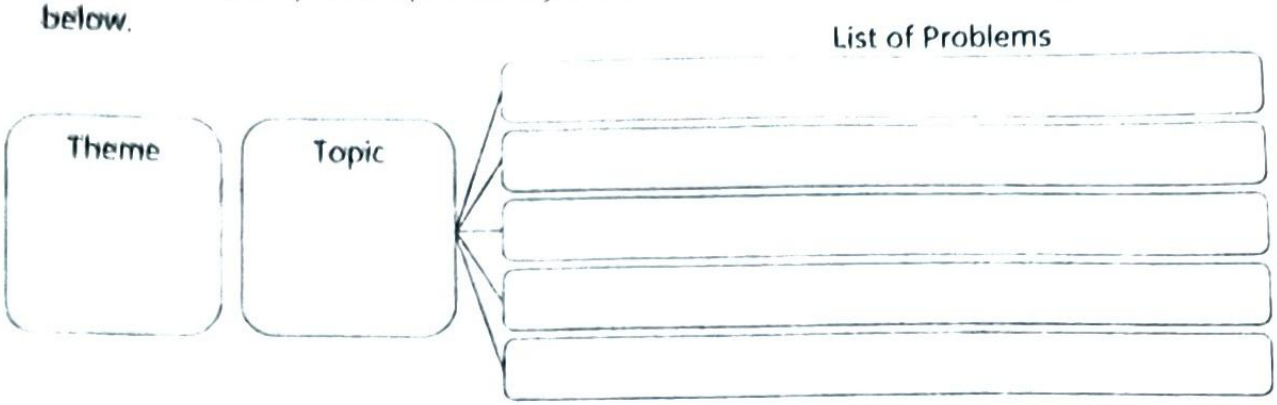
1. Think of a theme and write it in the space provided below. Also, mention the reason why you have selected it.

Your Selected Theme	Reason

2. Now, identify problems related to your theme and write them in the space provided below.



3. Prepare a mind map on the problems you have found and write them in the space provided below.



Goal of a Project

One of the most important steps in the project planning process is defining the project's goals. All subsequent actions are guided by a clearly defined aim, which offers clarity and direction. The process of defining the goal of an AI project involves iterations and calls for cooperation, research, and careful evaluation of numerous elements. A well-defined goal is the foundation of the success of a project by guiding the decision-making, ensuring the complete AI solution that addresses the real-life challenges.

4Ws Problem Canvas

Problem scoping refers to identifying a problem, finding out the factors that affect it, and defining the goal of the project. 4Ws Problem Canvas helps us in this process. It provides a structured framework for analysing and understanding a problem. It consists of four crucial parameters that we need to know while solving a problem: Who? What? Where? Why?



Who

The "Who?" block helps identify the people who are directly or indirectly affected by the problem. These people are referred to as stakeholders. Stakeholders are involved in the problem and will benefit from the solution we arrive at.

Activity 11.3

Fill in the "Who?" canvas based on the problem you have shortlisted for the activity.

Who?

1. Who are the stakeholders?



2. What do you know about them?

What

The "What?" block helps determine if the problem really exists. It helps to clearly define the nature of the problem. Under this block, you can gather evidence, like observations, newspapers, social media posts, reports, etc.

Activity 11.4

Fill in the "What?" canvas to determine the nature of your problem.

What

1. What is the problem?



2. How do you know that it is a problem? Is there any evidence?

Where?

Now, we need to know the context or location of the problem. The "Where?" block helps identify the situations where the problem is observed or has an impact. These could include physical locations, like cities or specific areas, online platforms or contexts, such as educational institutes or public spaces, or whether it has a global or local impact.

Activity 11.5

Fill in the "Where?" canvas to determine the locations where it is prominent.

Where?

1. What is the context or location where the stakeholders experience the problem?



2. Where is the problem located?

Why

In the "Why?" canvas, think about why the given problem is worth solving and how the so... would benefit the stakeholders as well as society. This could include a better quality of li... increase in efficiency, cost savings, enhanced safety, long-term sustainability, or other po... outcomes.

Activity 11.6

Fill in the "Why?" canvas to determine the benefits that stakeholders would rece... from the proposed solution and how it would help them and society.

Why?

1. Why will this solution be of value to the stakeholders?



2. How will the solution improve their situation?

Thus, the 4W's problem canvas helps to define the problem, identify the stakeholders, specify... context, and highlight the importance of solving the problem.

Problem Statement Template

The Problem Statement Template aids in summarising all the key points in the 4Ws problem ca... into a single template, which enables us to quickly get back the ideas as needed in the future. For... purpose of further analysis and decision-making, this template makes it simple to understand... remember the important aspects of the problem.

Implementing the 4Ws Problem Canvas and Problem Statement Template

Let us see how to use the '4Ws Problem Canvas' and summarise the key points with the 'Probl... Statement Template'. Consider the following scenario:

The city of Gopalapuram is a small urban area known for its clean and green environm... However, with rapid industrialisation, construction activities, and an increase in vehicular tra...

2. Where is the problem located?

_____

_____

_____

Why

In the "Why?" canvas, think about why the given problem is worth solving and how the so would benefit the stakeholders as well as society. This could include a better quality of li increase in efficiency, cost savings, enhanced safety, long-term sustainability, or other po outcomes.

Activity 11.6

Fill in the "Why?" canvas to determine the benefits that stakeholders would rece from the proposed solution and how it would help them and society.

Why?

1. Why will this solution be of value to the stakeholders?



_____

_____

_____

2. How will the solution improve their situation?

_____

_____

_____

Thus, the 4Ws problem canvas helps to define the problem, identify the stakeholders, specify context, and highlight the importance of solving the problem.

Problem Statement Template

The Problem Statement Template aids in summarising all the key points in the 4Ws problem can into a single template, which enables us to quickly get back the ideas as needed in the future. For purpose of further analysis and decision-making, this template makes it simple to understand remember the important aspects of the problem.

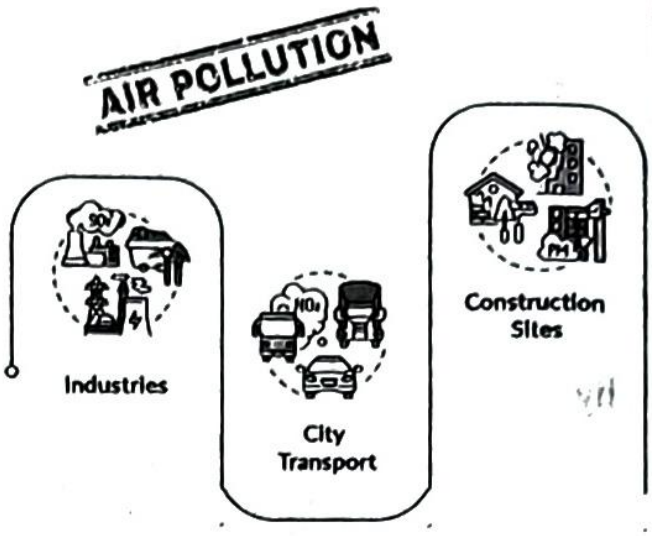
Implementing the 4Ws Problem Canvas and Problem Statement Template

Let us see how to use the '4Ws Problem Canvas' and summarise the key points with the 'Prob Statement Template'. Consider the following scenario:

The city of Gopalapuram is a small urban area known for its clean and green environme However, with rapid industrialisation, construction activities, and an increase in vehicular tra

ɔcal authorities have recorded an increase in ollution. Health authorities have reported a n respiratory problems and other health issues ıg citizens.

environmentalists have observed a decrease ısect and bird populations, disturbing the ıtc balance of the city's ecosystem. Let us fill e 4W's problem canvas and Problem Statement ılate for this scenario.



; Canvas

Who?

Who are the stakeholders?

Residents of Gopalapuram, local government authorities, environmentalists, healthcare uthorities, animals and birds, and industrialists.

What do you know about them?

- Residents are the primary stakeholders as their health and quality of life are directly affected by polluted air.
- Local government authorities are responsible for implementing policies and regulations related to environmental protection and public health.
- Environmentalists work for environmental protection and preservation of biodiversity.
- Health authorities plan and provide medical care for individuals affected by health issues.
- Industrialists wish to maintain a sustainable environment as part of their corporate social responsibility.

What?

What is the problem?

- Increase in air pollution caused by industrial emissions, vehicular pollution, and other sources.
- Increase on respiratory problems and other negative health effects in the citizens.
- The air pollution is affecting the insect and bird populations in the area and disturbing the ecosystem.



How do you know that it is a problem?

- Local authorities have recorded an increase in air pollution.
- Health authorities have reported a rise in respiratory problems and other negative health effects. Environmentalists have observed a decrease in insect and bird populations.

Where?

5. What is the context or situation in which the stakeholders experience the problem?

   The impact of air pollution is felt in different locations and in the daily lives of the residents.

6. Where is the problem located?

   Residential areas, public places, workplaces, green spaces, educational institutions, and natural habitats.



Why?

7. Why will this solution be of value to the stakeholders?

   - If air quality improves, there will be reduction in air pollution and health risks for residents.
   - There will be reduction in the healthcare burden on the health authorities.
   - This will also help industrialists and authorities ensure sustainable development.
   - Environmentalists can be assured of the preservation of biodiversity and green spaces.



8. How will the solution improve their situation?

   - Citizens will experience a cleaner and more enjoyable living environment.
   - If air quality improves, there will be reduction in air pollution and health risks for residents.
   - It will contribute to building a sustainable and environmentally conscious city.
   - It will help maintain ecological balance by promoting a healthy environment for the insect and bird population.

Data Acquisition

Data acquisition is the second stage of the AI Project Cycle. As the term clearly mentions, data acquisition is the process of acquiring or collecting accurate and reliable data from relevant sources. Let us first understand what data is.

What is Data?

Data is a collection of facts or statistics collected for reference or analysis. Data can be processed into meaningful information using various data analysis techniques and algorithms. Data can be in the form of text, video, images, audio, and so on.

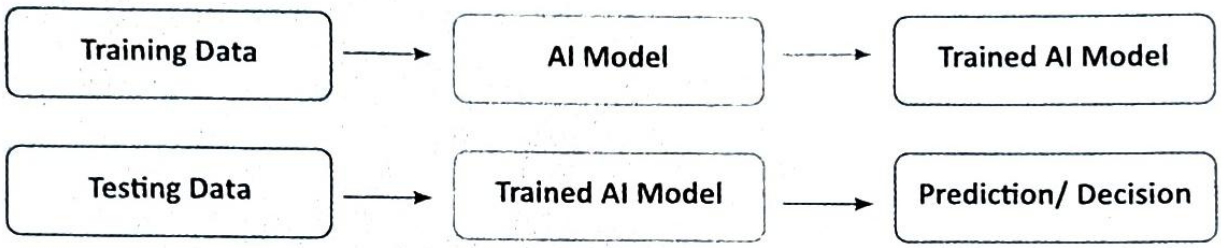
Data is the fuel for AI algorithms. When we want AI systems to make predictions, we need to use data to train them. AI algorithms learn and work based on patterns and insights derived from data.

Types of Data

Data can be classified into the following categories while developing an AI project:

Training data: Training data is the initial dataset used to train an AI model. It is a set of examples that helps the AI model learn and identify patterns or perform particular tasks. We must ensure that the data used to train the AI model is aligned with the problem statement and is sufficient, relevant, accurate, and wide-ranging.

Testing data: Testing data is used to evaluate the performance of the AI model. It is the data that the AI algorithm has not seen before. It allows to check the accuracy of the AI model. Testing data should represent the information that the AI model will encounter practically in real-world situations.



Let us consider an example where we try to predict the result of a cricket match between **Team A** and **Team B** based on pitch conditions. In this example, the data used for training will include the pitch conditions for **Team A** and **Team B** in previous matches, along with the match outcomes.

The AI model learns about the relationship between the pitch conditions and the match results during training. Testing data will consist of the pitch conditions for upcoming matches. The AI model then uses this testing data to predict the most likely match outcome based on the given pitch conditions.

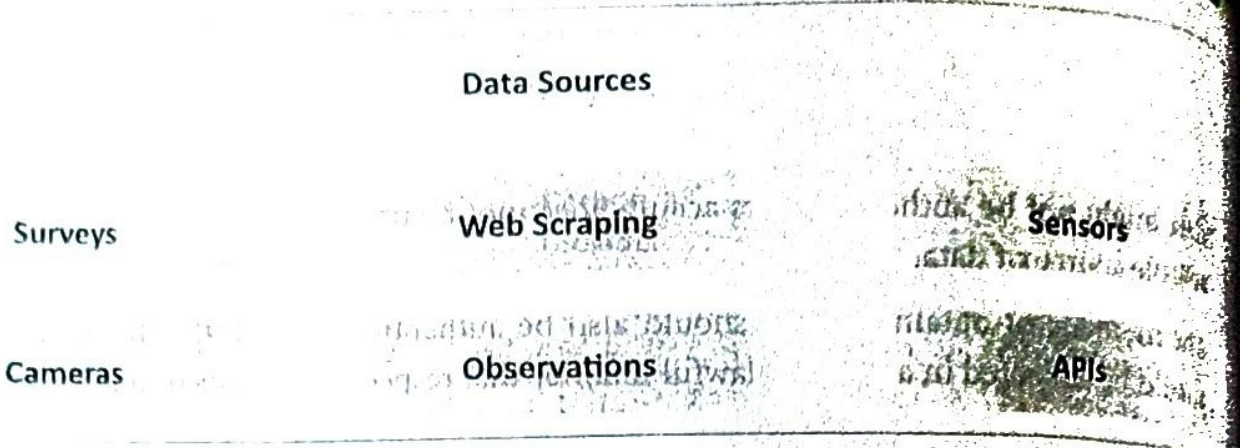
Data Features

Data features describe the type of information that will be collected in response to the problem statement. In order to determine the required data, we first need to visualise the factors influencing the problem statement. For this, we need to extract the data features for the problem scoped and find out the parameters that affect the problem statement directly or indirectly.

For instance, in the example used above for prediction of the cricket match outcome, the data features will be historical pitch conditions, scores, and match outcomes during matches played between Team A and Team B, and the pitch conditions of the upcoming matches.

Sources of Data

Once the data features have been shortlisted, we need to find sources to acquire the required data. Some of the sources of data are:



Surveys: A survey is a method of gathering specific information from a group of people by asking them questions. This enables us to collect valuable data quickly and efficiently. Surveys can be conducted on paper, through face-to-face or telephone interviews, or through online forms. For example, population census surveys are conducted once every ten years for population analysis.

Web scraping: Web scraping is the process of collecting information from websites in an automated manner. You can copy the data from a website, paste it in a document, and use it as a data source. Web scraping can be done manually, but if a large number of websites need to be accessed, some automated web scraping tools, like BeautifulSoup or Scrapy, can be used to help speed up the process.

Sensors: Sensors are devices that detect and measure environmental conditions, such as temperature, pressure, light, sound, and motion. They convert these physical parameters into electrical signals or digital data that can be processed and analysed by AI systems. They are often integrated with IoT devices. For example, in a smart home, sensors can be used to measure the room temperature or in autonomous cars to detect objects in the surrounding area.

Cameras: Cameras such as cell phone cameras, webcams, CCTV, and surveillance systems can be used to capture visual information in the form of images and videos. They play an important role in the Computer Vision domain of AI by analysing and understanding visual data. For example, capturing real-time videos of the surroundings through surveillance cameras.

Observations: It involves gathering data manually through direct observations of real-world events as they happen. For example, counting the number of birds in an area in a particular season.

APIs: Application Programming Interfaces (APIs) are programs used by developers to acquire data from other programs, services, or databases to extract relevant data required for the AI project. For example, if there is an AI project involving sentiment analysis, developers can use a social media API to access user posts or comments on Twitter or Facebook. Here, data acquisition is done automatically through special programs.

Kaggle platform: Kaggle is a popular platform for data science and machine learning datasets. It includes a lot of interesting datasets on all sorts of topics, such as text analysis, image recognition, tabular data, etc. It also hosts machine learning competitions and offers an integrated notebook environment that supports Python, R, and other popular data science languages.

portance of Acquiring Relevant Data from the Reliable Sources

Data forms the basis of the AI project; hence, data has to be authentic, reliable, and accurate.

The internet can be used to acquire data for your project from some random websites, but such data might not be authentic as its accuracy cannot be proved. Hence, it is necessary to find a reliable source of data.

The methods of obtaining data should also be authentic and ethical, which means the data should be collected in a fair and lawful manner that respects the rights of others. This will help avoid conflicts or negative consequences in the future.

Data should be open-sourced and not somebody's property. Also, accessing private data without permission is an offence.

One of the most reliable and authentic sources of information is the open source websites hosted by the government. These government portals have information presented in a suitable format that can be downloaded and used. Some of the open source government portals are: gov.in and, india.gov.in.

tem Maps

em maps are visual diagrams that help to see and understand the different parts or elements of AI project. They show how all the elements are connected or related to each other. They can be to understand the system's boundaries and how it interacts with elements in its surroundings.

ave listed the data features of the AI project. The system map will help us find the relationship een the elements of the problem we have scoped. It also helps us plan and arrive at a solution hieve our goal.

e important parts of a system map are listed below:

ents: These are the main parts of the system map that work together to achieve a specific purpose.

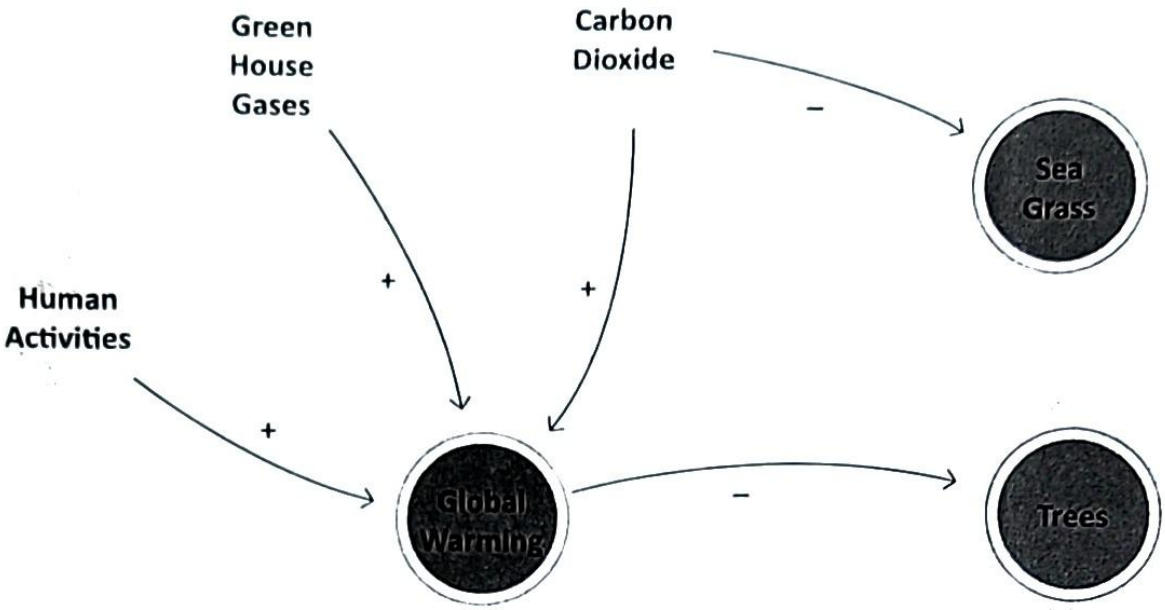
tionships or interconnections: These reflect the relationship between the different elements e system.

dback loops: Loops are specific chains of cause and effect where the output of the system ences its own behaviour or inputs. Feedback loops can be positive or negative, depending on ther they increase or decrease the effects within the system.

following is an example of a system map of a real-world problem: global warming. Global ming refers to the long-term increase in Earth's temperature due to human activities, mainly the ssion of greenhouse gases into the atmosphere. The primary elements associated with global ming are Human activities, Greenhouse Gases, Carbon Dioxide (CO_2) Emissions, etc.

he given system map, all the elements (X) of the global warming (Y) cycle are put in circles. map here shows the cause-and-effect relationship of elements with each other with the help rrows. The arrowheads depict the direction of the effect, and the sign (+ or -) shows their tionship. If the arrow goes from X to Y with a positive (+) sign, it means that both are directly ted to each other. That is, if X increases, Y also increases, and vice versa. On the other hand, if

the arrow goes from X to Y with a negative (-) sign, it means that both elements are inversely related to each other, which means if X increases, Y will decrease, and vice versa.



For example, the "Human Activities" element represents a positive loop on the system map of global warming. This is due to the fact that human activities like deforestation and the burning of fossil fuels emit greenhouse gases (like CO_2) into the atmosphere, which cause an increase in the Earth's average temperature, which is known as global warming.

A negative loop is shown by the "Trees" element. This is because plants play a crucial role in the global Carbon cycle through a process called photosynthesis. During photosynthesis, plants absorb Carbon Dioxide (CO_2) from the atmosphere and convert it into organic matter while releasing oxygen; thus, removing CO_2 from the atmosphere. Furthermore, seagrasses can absorb and hold carbon up to 35 times faster than tropical rainforests, accounting for 10–18% of total ocean carbon storage. In such a way, seagrasses play a crucial role in reducing CO_2 levels and contributing to the prevention of global warming.

Data Exploration

After acquiring the data required for the system, we need to explore it. Data exploration is the process of analysing data to discover patterns and gain insights. In order to explore data, different data visualisation methods are used, like bar chart, line chart, histograms, etc.

Need for Visualising Data

Data visualisation helps us in the following ways:

- It simplifies complex data; thus, making it easier to comprehend.
- It helps gain a deeper understanding of the trends, relationships, and patterns present within the data.

- It uncovers hidden relationships or anomalies (odd behaviour) that may not be immediately apparent.
- It helps us in selecting models for the subsequent AI Project Cycle stage.
- It makes it easy to communicate insights to others, even non-technical people.

Data Visualisation Tools



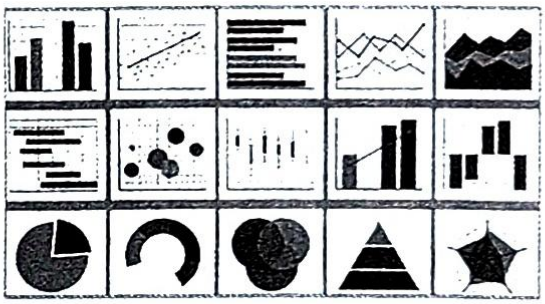
Data visualisation tools can be either manual or automated and can be selected based on the nature of the data and its complexity. There are various data visualisation tools as listed below:

Tableau: It is a popular data visualisation tool. It is a user-friendly and interactive tool that supports large datasets. It provides options to collaborate with others and securely share data from a variety of sources.

Microsoft Power BI: It is a cloud-based business intelligence tool that provides options to access data from multiple sources into a single dataset, make interactive dashboards, evaluate data, create reports, and share data with others.

Qlik View: It is a data visualisation tool that shows the relationship between each piece of data that is present in a dataset. Data is held in memory for multiple users; hence, it gives a super-fast user experience. The relationship between data is not shown by arrows or lines but by colours.

Looker Studio: It is a free and powerful web-based tool for creating interactive charts, dashboards, and reports. It offers easy tools to automatically combine data from a wide range of sources, analyse data, and enable you to collaborate with others in real time.

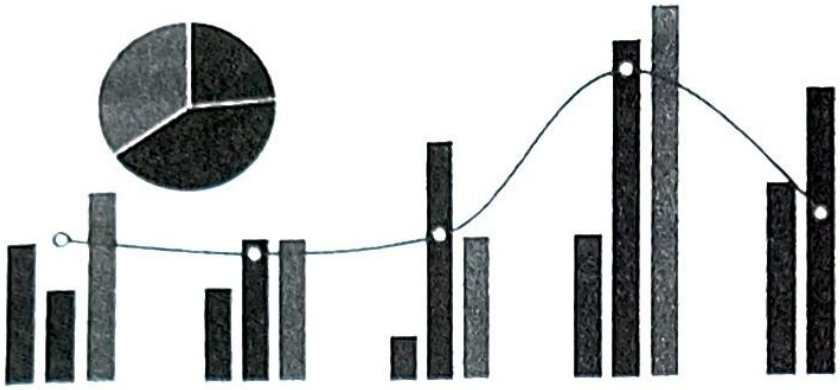


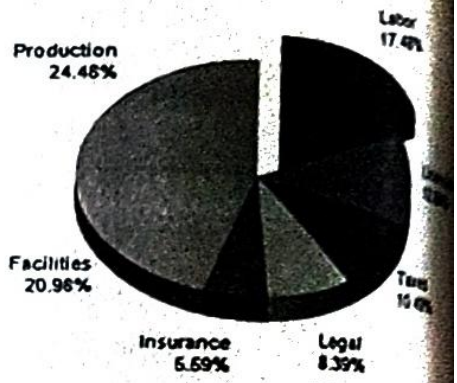
Excel: It offers data visualisation through graphs, charts, as well as templates. Excel is a manual visualisation tool that offers a range of charts, like column charts, bar charts, pie charts, line charts, area charts, scatter charts, etc. It lacks advanced features and cannot be used for large amount of data, but it offers a simple and familiar user interface.

Plotly: It is an open-source module of Python that uses JavaScript and supports interactive web-based 2D and 3D graphs, like line charts, 3D scatter plots, bar charts, area plots, etc. It is visually attractive and allows customisation of graphs, which makes them more meaningful and understandable.

Matplotlib: Matplotlib is a Python library used for creating static, animated, and interactive visualisations in Python. It can create high quality and customisable plots.

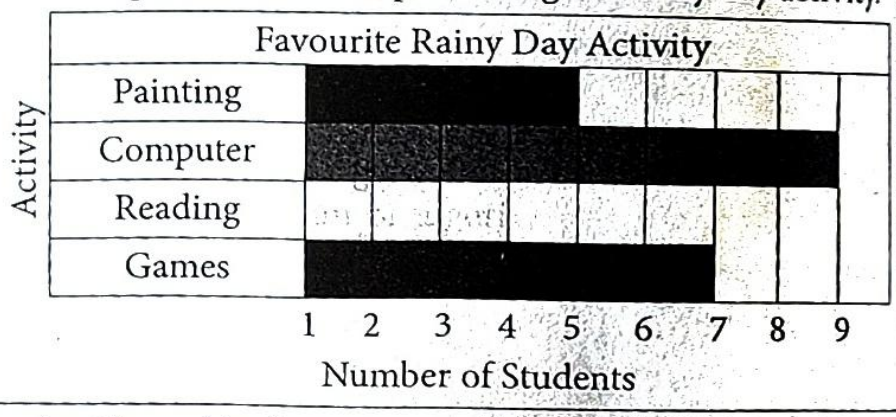
Highcharts: It is a JavaScript library and a free tool that supports touch-screen-based platforms, like Android and iOS. It is dynamic and customisable, includes a range of chart types, and supports integration with web applications.



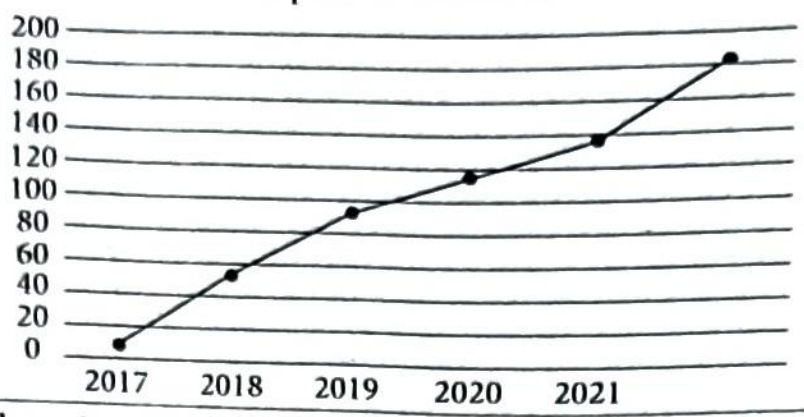


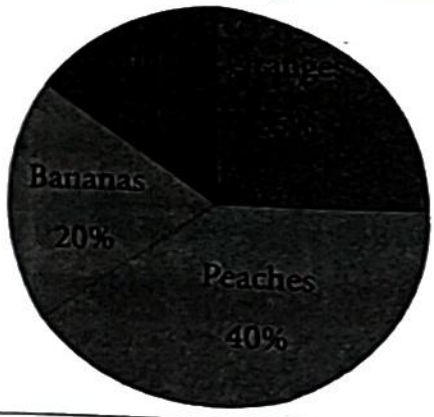
Ways to Visualise Data

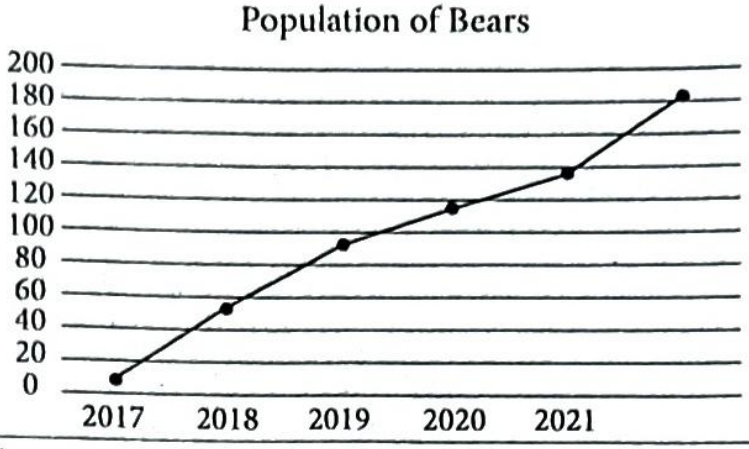
There are different techniques using which you can represent your data visually. Let us learn a them in detail.

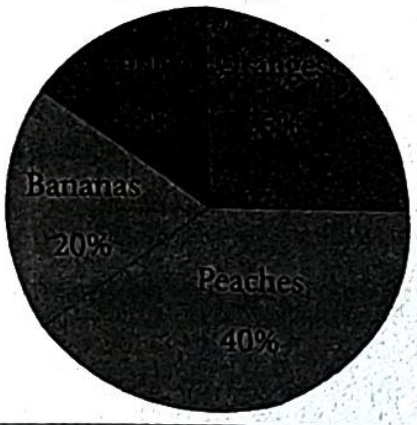
Data Visualisation Technique 1	
Name of the Representation	**Bar Chart**
One-line Description	A bar chart is a chart that presents categorical or grouped data w rectangular bars, where the height or length of the bars is proportio to the values that they represent.
How to draw it	The following is a bar chart representing the rainy day activity. 
Suitable for which data type?	A bar chart is used to show comparison between groups of categori of data. The above graph draws a comparison between the numb of students who enjoy a particular activity on a rainy day. Some othe examples are the scores of different teams in a cricket tournament o the marks of students in different subjects.

Data Visualisation Technique 2	
Name of the Representation	**Line Chart**
One-line Description	A line chart is a chart that is created by plotting a series of points th are connected with the help of a line. It is used to track changes i values over a period of time.

low to draw it	The following is a line chart representing the population of bears. 
Suitable for ich data type?	A line chart is suitable for displaying quantitative values over a specified time interval to identify trends in the data. The above chart shows the change in the bear population in a forest every year over a number of years. Other examples are variations in the temperature and population over a number of years.

isualisation Technique 3	
Name of the presentation	**Pie Chart**
ine Description	A pie chart represents data in a circular form, divided into sectors, where the size of each sector is proportional to the quantity it represents. It helps to understand the contribution of individual components or categories to the whole.
w to draw it	The following is a pie chart representing the favourite fruit of students. 
itable for h data type?	A pie chart is suitable for displaying data that can be divided into different categories. It is commonly used to show the proportional or percentage distribution of different categories within a whole. The above chart shows the percentage of each type of fruit preferred by students within the broader category of fruits. Other examples of using a pie chart include displaying the proportion of different sources of energy used in a country or the proportion of monthly expenses by expense category.

How to draw it	The following is a line chart representing the population of bears. 
Suitable for which data type?	A line chart is suitable for displaying quantitative values over a specified time interval to identify trends in the data. The above chart shows the change in the bear population in a forest every year over a number of years. Other examples are variations in the temperature and population over a number of years.

Visualisation Technique 3	
Name of the Representation	**Pie Chart**
One-line Description	A pie chart represents data in a circular form, divided into sectors, where the size of each sector is proportional to the quantity it represents. It helps to understand the contribution of individual components or categories to the whole.
How to draw it	The following is a pie chart representing the favourite fruit of students. 
Suitable for which data type?	A pie chart is suitable for displaying data that can be divided into different categories. It is commonly used to show the proportional or percentage distribution of different categories within a whole. The above chart shows the percentage of each type of fruit preferred by students within the broader category of fruits. Other examples of using a pie chart include displaying the proportion of different sources of energy used in a country or the proportion of monthly expenses by expense category.

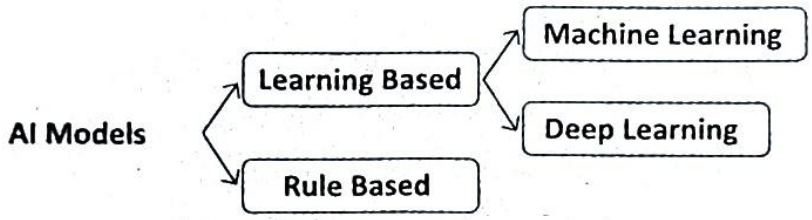
For example, predicting the weather forecast for the next seven days based on data from the previous year and the previous week.

Deep learning: Deep learning enables machines to learn from and perform tasks on large amount of data or big data. Due to the large amount of data, the system learns on its own by using multiple machine learning algorithms working together to perform a specific task. Deep learning is based on neural networks with multiple layers that mimic the working of the human brain.

Deep learning models understand how the human brain works in different situations and then try to recreate that behaviour. For example, a deep learning model in a driverless car can identify a person crossing the road.

Types of AI Models

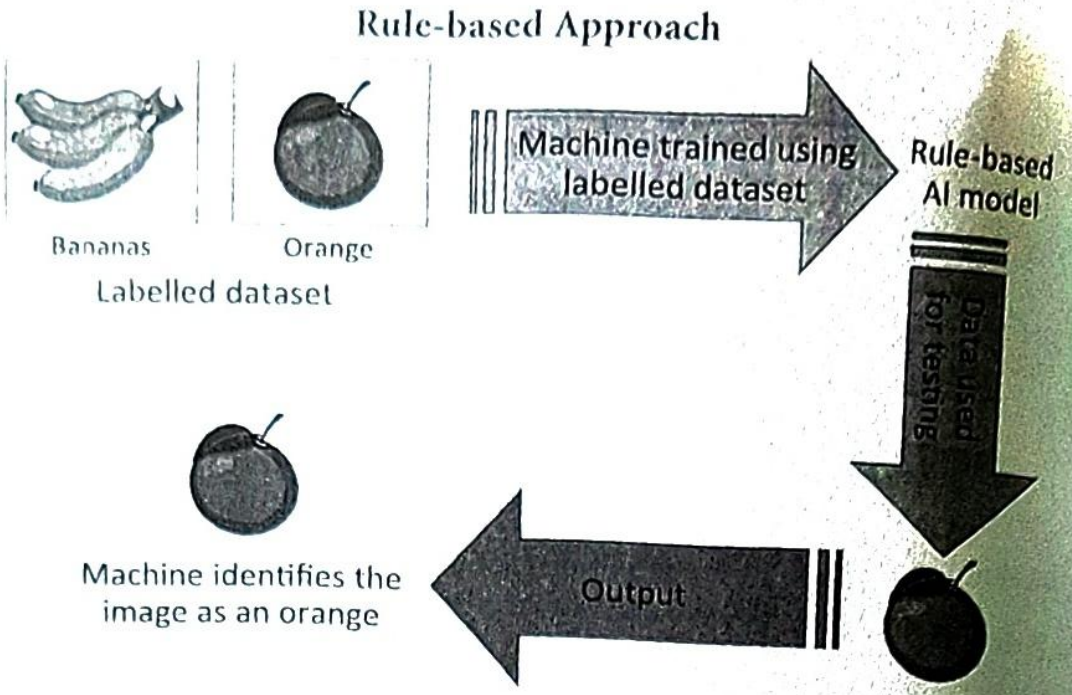
In the data exploration stage, data is represented graphically so that it can be made easily understandable to humans and we can discover the patterns and trends in data. However, in order to understand and analyse data as well as find patterns and trends in it, machines need data in the form of binary numbers, i.e., 0s and 1s. AI models define the relationship between different parameters mathematically. Hence, developing AI models involves using a mathematical approach to analyse data. AI models are classified into the following approaches:



Rule-based Approach

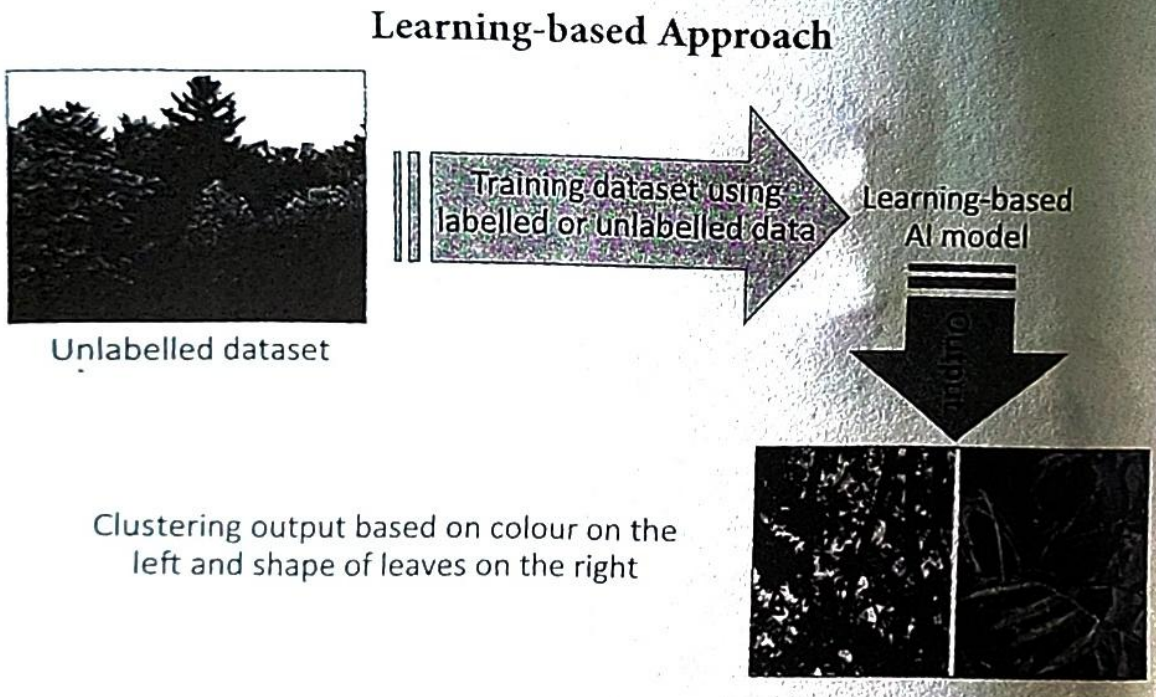
Under the rule-based approach, the developer feeds data along with some rules to the model. The rules are in the form: "If this happens, then do this". In other words, a rule-based approach refers to AI modelling where the relationships or patterns in the data are defined by the developer. The machine follows the rules defined by the developer and produces the required output. Let us assume that you want a machine using an AI model to be able to correctly identify and differentiate between the images of bananas and oranges. If you have a dataset of 200 images of bananas and oranges, you will first have to train the machine by feeding it images of the fruits labelled appropriately as "banana" and "orange" so that it can easily identify them. After the training, if you test the machine with the image of an orange, it will compare the image with the data and labels of the images used during training and identify the test image as an orange. The rules given to the machine, in this case, are in the form of labels provided for each image during the training.

In this approach, if the machine is shown any data for which no rules have been defined, it will not be able to understand or identify the data. Consider the aforementioned example. If apples were presented to the machine instead of bananas or oranges, it would not be able to recognise them.



Learning-based Approach

In the learning-based approach, the machine learns on its own. The data used in this model is both labelled and unlabelled. The machine looks at the data, tries to find similar patterns in it, and groups data that is alike. The relationships or patterns in the data are not defined by the developers and the AI model define its own set of rules. As an output, the machine communicates the trends it observed in the training data. For example, suppose you have a random mix of 1000 images of different trees in a forest area, and you do not know any particular trend in the data since you are unaware of the height, colour, or any other features. If this data is fed into a learning-based AI machine, the machine would identify the trends or patterns followed in the features of these 1000 images. It will then classify or cluster these images based on their height, colour, fruit, types of leaves, etc.



Clustering output based on colour on the left and shape of leaves on the right

The following are the differences between rule-based and learning-based approaches:

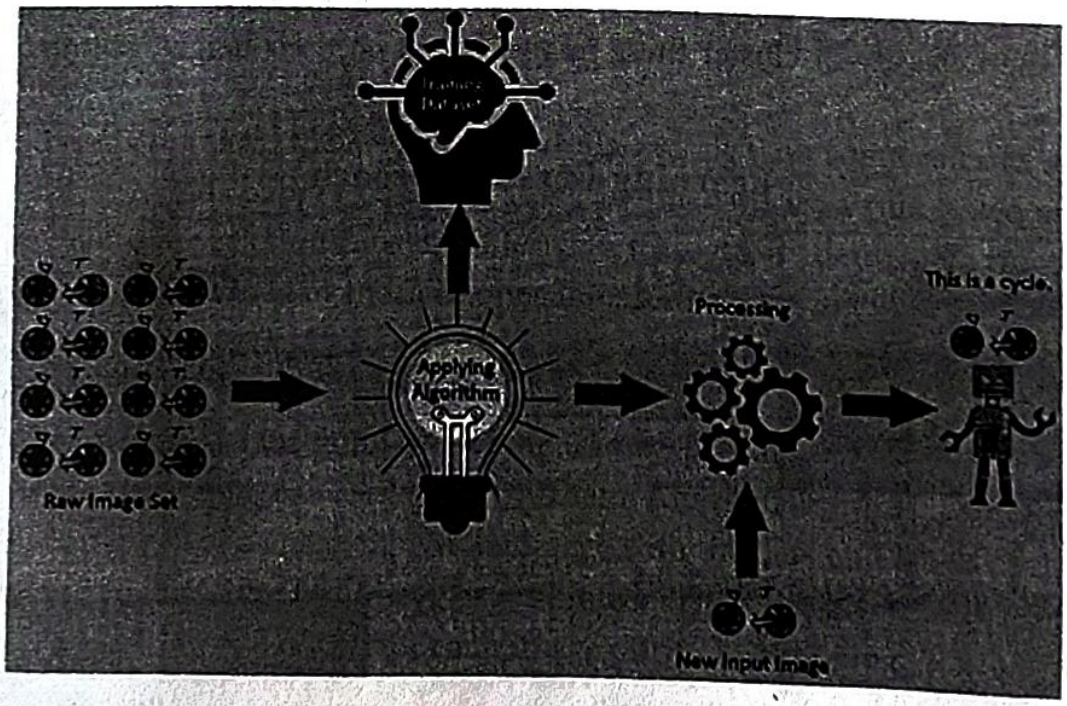
Rule-based Approach	Learning-based Approach
The machine follows the rules defined by the developer.	The machine learns on its own from the data.
AI is achieved through rule-based technique.	AI is achieved through learning technique.
It typically uses labelled data.	It can handle both labelled and unlabelled data.
It may require less training time.	It requires more training time.

Supervised Learning

In supervised learning, the model is trained on a labelled dataset. It means that we have both raw input data as well as output. We divide the data into a training dataset and test dataset. The training dataset trains the model while the test dataset acts as new data for predicting the output or to see the accuracy of the model. In this type of learning, the model learns from the known results. Accuracy is what we achieve in supervised learning.

This model performs at a fast pace because the time taken for the training is less. After all, the result is already known to us. This model produces accurate results on unseen data or new data without even knowing a previous target. The algorithm understands the input patterns that produce the expected output. Once the algorithm is trained, it predicts the correct result from an entirely new input.

In the image below, we see that an image of a cycle is fed into the algorithm as a raw input. There is a supervisor who keeps on correcting and training the machines. The supervisor keeps telling "yes, it is a cycle" or "no, it is not a cycle." This process is repeated until a final trained model is produced. Once the model is produced, it can easily predict the correct output of an unknown input.

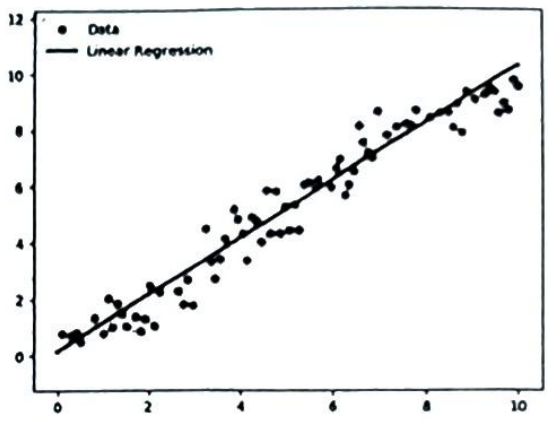


Regression

Regression is a type of supervised learning model used to predict continuous numerical values. Continuous data means data that can have any value within a certain range, for example, the price of a product or the salary of an employee. It is used to predict the behaviour of one variable depending on the value of another variable.

A real-life example of regression is predicting the temperature for the next day based on historical weather data, including factors like humidity and cloud cover.

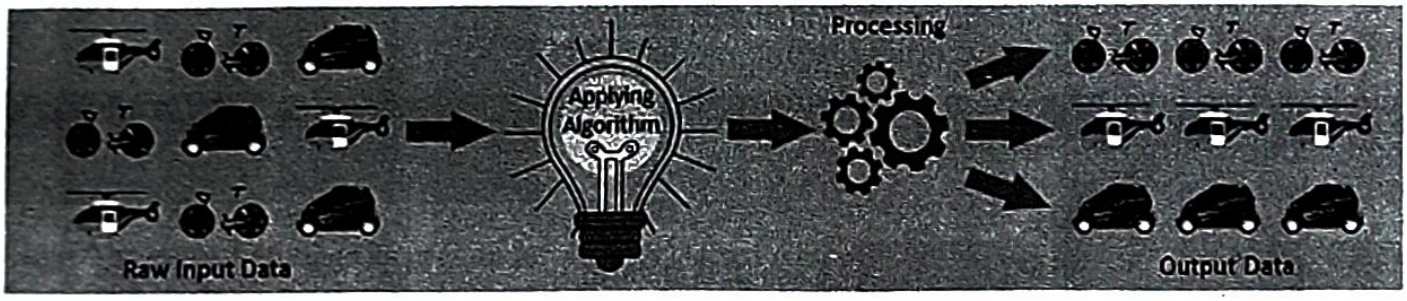
A special function called mapping helps us predict their future values based on the data we have. For example, we can use an employee's past salary data to train the AI model to predict the future salary.



The algorithm learns from this data and creates a solid line on a graph that represents the pattern in the salaries. In the given graph, the blue dots represent the data values we have, i.e., the past salaries, and the solid line represents the function that helps us predict future values.

Unsupervised Learning

In this type of learning, the input used to train is neither classified nor labelled. This learning studies on how systems can understand a function to describe a latent structure from the unlabelled data. The chief function of unsupervised learning is to understand the patterns hidden in the data. Once a model learns to understand the patterns, it can predict the patterns for any new dataset. The system does not understand the correct output, but explores the data and can draw inferences from the given datasets to describe the latent structures from any unlabelled data.



We know that in unsupervised learning, our dataset is not labelled. If we feed the images of a cycle, car, and helicopter as raw inputs, our model will easily differentiate among the three. However, it will not be able to tell whether a given cluster is of a cycle or not as it is unlabelled.

AI Project Evaluation

Evaluation is the process of assessing the reliability and efficiency of any AI model based on the output generated by feeding the model with the testing data and comparing it with the expected outcome. Once a model has been developed and trained, it is important to thoroughly test its performance and calculate its efficiency. In order to achieve this, the model is tested using a dataset, called testing data, that was separated from the acquired dataset at the data acquisition stage.

The efficiency and effectiveness of the model are determined on the basis of various parameters that help assess its overall performance. Evaluation happens at various stages of the AI project cycle to identify areas for improvement and make necessary adjustments in order to achieve the project goals.

Prediction

A key component of AI evaluation is Prediction since it measures how well a machine learning or AI model performs on a particular job. The process of making forecasts or inferences about unknown or future data using a trained model and the patterns it has learned from training data is known as prediction. There are numerous new terms that we take up for the task of evaluating a model. Let us acquaint ourselves with these keywords with the help of an example.

The Scenario

Suppose, we use an AI-based prediction model that predicts the rainfall at a particular place. The objective of the model is to predict whether it is raining at a particular place or not.

In order to understand the effectiveness of this model, it is necessary to see if the predictions that it makes are correct or not. There are two conditions that we need to consider: Prediction and Reality. Prediction is the output produced by a machine and Reality is the actual scenario in that particular place when the prediction has been made. Let us consider the different combinations that we can experience in these two conditions.

Case 1: Is it raining?

PREDICTION: YES	REALITY: YES	TRUE POSITIVE



We can clearly see in the picture that it is in fact raining. The model predicts a Yes. According to the model, it is raining. The prediction is matching the reality. This condition is called the **True Positive**.



Case 2: Is it raining?

PREDICTION: NO	REALITY: NO	TRUE NEGATIVE

There is no rain at the given place, and therefore the reality is no. the model has also predicted it correctly as a no. here, the prediction is matching the reality. This condition is called the **True Negative**.

Case 3: Is it raining?

PREDICTION: YES	REALITY: NO	FALSE POSITIVE



The reality in this picture is that there is no rain, but the machine has incorrectly predicted that it is raining. There is a mismatch between the reality and prediction. This case is called the **False Positive**.

Case 4: Is it raining?

PREDICTION: NO	REALITY: NO	FALSE NEGATIVE



It is clear in this picture that it is in fact raining. The reality is a Yes, but the model has incorrectly predicted it as a No, which means that the machine predicts that there is no rain at the location. This condition is called the **False Negative.**

AI Project Deployment

AI project deployment is the process of implementing an AI model in a real-world scenario. model is integrated into the desired software or system and packaged in such a way that it c used for practical applications. The goal is to make the AI model useful for solving real-w problems.

ReCap

- The AI project cycle consists of five stages: problem scoping, data acquisition, data explora modelling, and evaluation.
- Problem scoping allows you to set clear goals and outline the objectives of the AI project.
- Data acquisition focuses on collecting relevant data required for the AI system.
- Data exploration involves the exploration and analysis of the collected data to interpret patter trends, and relationships.
- In the modelling stage, you can select the appropriate AI model to achieve the goal.
- The evaluation stage focuses on testing the selected AI model, and the results need to be compare with the expected outcome.
- The 4Ws problem canvas is a structured framework that helps in problem scoping.
- The Problem Statement Template aids in summarising all the key points in the 4Ws problem canv into a single template.
- Training data is the initial dataset used to train an AI model and testing data is used to evaluate the performance of the AI model.
- Data features describe the type of information that will be collected in response to the problem statement.
- System maps are visual diagrams that help to see the different parts or elements of the AI project.
- In the rule-based approach, the machine follows the rules defined by the developer and produces the required output.
- In the learning-based approach, the machine learns on its own.
- In unsupervised learning, the input used to train is neither classified nor labelled.